Venkata Lakshmi Parimala Pasupuleti

Senior Full Stack Engineer · Generative AI Integration · Regulated Data · Prototypes to Production · Python · React · TypeScript · AWS · 9yr · DMV

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PROFESSIONAL SUMMARY

Senior full stack engineer with 9+ years of production experience who moves between domains, learns new tools quickly, and turns loosely defined ideas into working software - the generalist profile Oddball Labs is looking for. I'm comfortable from prototype through production: at Magizhs I took a GenAI application from zero to production as sole technical owner in weeks, navigating an ambiguous problem space where requirements were still taking shape. AgentProbe and LeaseI show GenAI integration applied where it creates real value - not AI for its own sake, but retrieval architectures, agentic pipelines, and evaluation systems built to solve specific problems. I build across the full stack (Python/FastAPI backend, React/TypeScript frontend, AWS infrastructure) and have a strong grounding in security, privacy, and compliance on sensitive data: HIPAA-compliant RBAC on AI on FHIR, HUD affordable housing compliance with zero violations across 7 years at Entrata, and deterministic HITL guardrails in regulated decision workflows. I know when to prototype quickly and when to engineer for scale - and I can tell the difference. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Based in Arlington, VA.

TECHNICAL SKILLS

Generative AI & Full Stack Engineering: Python (9yr), TypeScript, React.js, FastAPI, LangGraph, LangChain, Generative AI, LLM Integration, RAG Pipelines, Agentic Systems, Tool Calling, Multi-Agent Orchestration, ChromaDB, Vector Databases, Embeddings, Semantic Search, Prompt Engineering, LLM Evaluation, Output Validation, HITL Guardrails, Anthropic API, OpenAI API, Ollama, MCP, AI/ML Integration

Backend, APIs & Data: FastAPI, Flask, REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, High-Throughput Systems, Performance Optimization, Reliability Engineering, Idempotency, Docker, SQLite

Cloud, Security & Engineering Excellence: AWS (S3, EC2, RDS, Glue, Lambda), Docker, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Regression Testing, HIPAA Compliance, Security Best Practices, Privacy-First Design, Accessibility, Observability, Structured Logging, Monitoring, Alerting, Design Documents, Technical Mentoring, Code Reviews, Agile/Scrum, Linux, Bash, Git

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - 0→1 GenAI Product Delivery in Ambiguous Problem Space

- Took a GenAI application from zero to production as sole technical owner - designed architecture, built the system, and shipped to production in weeks while the problem space was still taking shape; built evaluation pipeline, CI validation gate, HITL guardrails for regulated decisions, and observability infrastructure; applied generative AI where it created real value, not as a default.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS cloud infrastructure; applied Isolation Forest anomaly detection across 3 research pipelines; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems across multiple sections.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Full stack production systems** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ users and 1M+ property records; shipped 20+ full-stack modules (Python, JavaScript, PostgreSQL, RabbitMQ) across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with automated testing; AWS S3/EC2/RDS/Glue infrastructure; React.js + TypeScript frontend features.
- **Security, compliance & regulated data** - owned HUD affordable housing compliance pipelines (eligibility, rent calculations, regulatory imports) with zero compliance violations across 7yr ownership; 100% PHI compliance on AWS S3 data migration with Libsodium-encrypted pipelines (zero security incidents); built structured logging, monitoring, and alerting infrastructure for regulated production systems.
- **Technical leadership & mentorship** - wrote design documents for 20+ production modules; mentored 5+ engineers; 300+ peer code reviews; 200+ SQL query optimizations (30% performance gain); Python/Bash automation eliminating 80% of manual workload; built internal debugging tools reducing triage time 40%.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: GenAI Evaluation Platform - Exploratory Engineering into Production

GenAI · RAG · Agentic · Evaluation · Observability · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Generative AI evaluation platform built through exploratory engineering - started with a loosely defined goal (how do you measure whether an AI agent is getting better or worse?) and iterated to a production system. **GenAI integration where it adds real value:** multi-provider LLM routing (Anthropic, OpenAI, Ollama) with cost/latency tradeoffs; RAG pipeline with ChromaDB semantic retrieval; 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency). **Prototypes to production:** 15-endpoint FastAPI backend, React.js + TypeScript frontend, CI-integrated regression gate (GitHub Actions) blocking quality degradation before deployment. **Agentic architecture:** LangGraph orchestration with structured distributed tracing across all agent steps - observability built in, not bolted on. Zero regressions across 1,240+ test cases.

Leasel: GenAI in a Regulated Problem Space - Ambiguity to Architecture

GenAI · Regulated Domain · Agentic · HITL · RAG · Privacy

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

GenAI applied to a complex regulated domain - the kind of ambiguous problem space where requirements evolve and constraints are discovered, not handed down. Started with a real investigation that took a day due to invisible system dependencies; ended with a production architecture. **Security, privacy & accessibility on regulated data:** deterministic HITL guardrails where the LLM reasons, human or deterministic logic decides every regulated compliance call - privacy preserved by design, never auto-resolving protected eligibility decisions. **Full stack:** FastAPI backend, Streamlit Engineer Console, RAG retrieval with 100% citation correctness, MCP integration for live regulatory data. 93.6% precision, 87.9% F1 across 1,240 test cases - applied AI that demonstrably works.

AI on FHIR: GenAI Integration on AWS - HIPAA, Security & Privacy

HIPAA · AWS · GenAI · Full Stack · Security · Compliance

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack GenAI application on AWS with HIPAA compliance and accessibility requirements - the security, privacy, and compliance grounding on sensitive data that the Oddball Labs role explicitly requires. LLM-powered NLP translating natural-language queries (EN/ES) into FHIR API calls; AWS Lambda serverless backend with API gateway; HIPAA-compliant RBAC with full audit trail on every AI-assisted retrieval; React.js + TypeScript frontend; Docker containerization; complete observability stack. ~60% reduction in manual retrieval time - AI integration that delivers measurable value.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87 · DMV local

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017